

Building Digital Resilience Through Vibe Coding: A Content Analysis of AI-Assisted Development for Sustainable Digital Transformation

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Abstract

Approximately 70% of digital transformation initiatives fail, costing organizations an estimated \$2.3 trillion annually, with traditional software development (\$434K–\$2.3M per project) and scarce technical talent as fundamental bottlenecks. In February 2025, Andrej Karpathy coined “vibe coding” to describe an AI-assisted programming paradigm in which developers describe desired functionality in natural language and accept AI-generated code, often without full comprehension. Within twelve months, 87% of Fortune 500 companies adopted at least one vibe coding platform, and the term was named Collins Dictionary Word of the Year 2025. Yet no theoretically grounded academic study has investigated how vibe coding contributes to organizational digital transformation and resilience. This paper addresses that gap through a directed content analysis of 97 sources spanning academic literature, industry reports, developer community discourse, and platform documentation. Using an integrated lens combining Dynamic Capabilities Theory, the Technology–Organization–Environment framework, and Diffusion of Innovation theory, we identify a *resilience paradox*: vibe coding simultaneously strengthens organizational resilience through accelerated sensing, seizing, and transforming capabilities while introducing fragilities including security vulnerabilities (45% of AI-generated code), technical debt, and a “vulnerable developer” phenomenon affecting 63% of practitioners. We extend Sanchis et al. (2020) by situating vibe coding as the fourth paradigm

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in software-development automation—following CASE, the RAD/IDE/web-framework era, and low-code—and propose a tiered governance framework for sustainable adoption. We forecast a bifurcated diffusion trajectory: rapid adoption for non-critical applications and a persistent quality threshold for enterprise-grade systems through 2030.

Keywords: vibe coding, digital transformation, digital resilience, AI-assisted development, low-code, dynamic capabilities, content analysis

Highlights

- First theory-grounded analysis of vibe coding’s organizational impact (n=97).
- Identifies a “resilience paradox”: vibe coding both builds and breaks resilience.
- Frames vibe coding as the fourth paradigm in software-development automation.
- Proposes a tiered governance framework matched to application criticality.
- Forecasts bifurcated diffusion to 2030 with explicit observable signposts.

1. Introduction

Enterprises operate in an environment of accelerating complexity and disruption. The increasing intensity of global competition, rapid technological change, and volatile market conditions demand that organizations build digital capabilities at unprecedented speed (Sanchis et al., 2020; Sanchis and Poler, 2019). Digital transformation—the integration of digital technologies into all areas of business, fundamentally changing how organizations operate and deliver value—has become a strategic imperative rather than an option (Vial, 2019). Yet the track record is sobering: approximately 70% of digital transformation initiatives fail to achieve their objectives (Tabrizi et al., 2019; Hess et al., 2016; Mergel et al., 2019), costing organizations an estimated \$2.3 trillion annually in unrealized value (Everest Group, 2024). A fundamental bottleneck lies in software development itself. Traditional development remains expensive, with project costs ranging from \$434,000 for

15 small and medium-sized enterprises to \$2,322,000 for larger projects (Fryling, 2019; Clancy, 2014). Development cycles measured in months or years are increasingly misaligned with the pace at which organizations must respond to market shifts and competitive threats.

The quest to democratize and accelerate software development is not
20 new. In the late 1980s, Computer-Aided Software Engineering (CASE) tools promised to automate significant portions of the development lifecycle. The CASE market grew to \$12.11 billion by 1995, yet 73.5% of companies never adopted these tools (International Data Corporation, 1996; Iivari, 1996). Unrealistic expectations, conceptual gaps between tool designers and users, high
25 training costs, and the overestimation of productivity gains all contributed to CASE’s failure to achieve mainstream adoption (Iivari, 1996; Huff, 1992). Two decades later, low-code development platforms (LCDPs) emerged as a more accessible alternative. Coined by Forrester Research in 2014 (Richardson and Rymer, 2014), “low-code” describes platforms that enable applica-
30 tion development through visual drag-and-drop interfaces, pre-built modules, and minimal hand-coding. Sanchis et al. (2020) demonstrated that low-code platforms can serve as enablers of digital transformation in manufacturing, offering rapidity (5–10× faster development), cost reduction, citizen developer involvement, and simplified maintenance. However, their analysis also
35 revealed significant limitations: scalability constraints for enterprise-grade applications, fragmentation across vendors, restriction to software-only systems, and substantial vendor lock-in (Sanchis et al., 2020; Tisi et al., 2019).

In February 2025, a new paradigm emerged that would fundamentally reshape this trajectory. Andrej Karpathy, co-founder of OpenAI and former
40 Director of AI at Tesla, posted on X (formerly Twitter): “There’s a new kind of coding I call ‘vibe coding,’ where you fully give in to the vibes, embrace exponentials, and forget that the code even exists” (Karpathy, 2025). Karpathy described a practice of using large language models (LLMs) to generate entire codebases from natural language prompts, accepting the out-
45 put largely without detailed review, and iterating through error messages by copying them back into the AI. Within twelve months of this coinage, vibe coding catalyzed a paradigm shift of remarkable velocity. The term was named Collins Dictionary Word of the Year 2025 (Collins Dictionary, 2025). Industry surveys reported that 92% of U.S. developers use AI coding tools
50 daily, with 62% reporting productivity gains of 25% or more (Stack Overflow, 2025). Gartner predicted that 60% of new code would be AI-generated by 2026 (Gartner, 2025). Perhaps most strikingly, 87% of Fortune 500 compa-

nies adopted at least one vibe coding platform, and 25% of Y Combinator’s Winter 2025 cohort reported running codebases that were 95% AI-generated (Shmargad, 2025; Y Combinator, 2025).

Despite this extraordinary adoption velocity, there is a significant gap in academic understanding. While substantial research exists on low-code platforms and digital transformation (Sanchis et al., 2020; Sahay et al., 2020; Cabot, 2020), AI coding tools and developer productivity (Ziegler et al., 2022; Imai, 2022), and organizational resilience frameworks (Sanchis and Poler, 2019; Browder et al., 2019), there is virtually no theoretically grounded academic research investigating how vibe coding contributes to digital transformation and organizational resilience across business sectors. Existing vibe coding research—primarily arXiv preprints and a grey literature review accepted at ICSE 2026 (Fawzy et al., 2025)—focuses overwhelmingly on individual developer productivity metrics and code quality, not on organizational-level resilience outcomes or cross-industry digital transformation impacts.

This paper addresses this gap through a directed content analysis (Hsieh and Shannon, 2005) investigating how vibe coding enables digital transformation and builds digital resilience across business sectors. We extend Sanchis et al.’s (2020) low-code/manufacturing framework to the vibe coding era and broader business contexts, providing the first theoretically grounded, cross-industry analysis of vibe coding’s organizational impact. Our analysis is structured through an integrated theoretical lens combining Dynamic Capabilities Theory (Teece et al., 1997; Warner and Wäger, 2019), the Technology-Organization-Environment (TOE) framework (Tornatzky and Fleischer, 1990), and the Diffusion of Innovation (DOI) Theory (Rogers, 2003), enabling us to examine not only what vibe coding enables but also how it diffuses across organizations and what organizational capabilities it enhances or diminishes.

Specifically, this study addresses three research questions:

RQ1: What is the current state of discourse on vibe coding as a paradigm for enabling digital transformation across business sectors?

RQ2: How does vibe coding contribute to (or hinder) organizational digital resilience compared to traditional low-code/no-code approaches?

RQ3: What are the enabling factors, barriers, and sustainability implications of vibe coding adoption for digital transformation?

By answering these questions, this paper makes three primary contributions. First, it provides the first systematic, multi-source content analysis of
 90 vibe coding’s organizational-level impact, synthesizing fragmented discourse across academic, industry, practitioner, and platform documentation sources. Second, it introduces the concept of the “resilience paradox”—the finding that vibe coding simultaneously strengthens and weakens organizational digital
 95 resilience—and theorizes this tension through the lens of Dynamic Capabilities. Third, it offers practical governance frameworks and a forecasting analysis spanning the 2025–2030 horizon, identifying critical transition factors and the boundary conditions under which vibe coding’s resilience-enhancing effects are likely to dominate its resilience-undermining effects, addressing recent calls for contingency-aware digital transformation theory (Hanelt et al.,
 100 2021; Verhoef et al., 2021).

2. Background and Theoretical Framework

2.1. *The Evolution of Software Development Automation*

The automation of software development has followed a punctuated evolutionary trajectory, with each paradigm shift promising to democratize development, reduce costs, and accelerate delivery. Understanding this historical evolution is essential for contextualizing vibe coding’s emergence and forecasting its trajectory.
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2.1.1. *CASE Tools (1980s–1990s)*

Computer-Aided Software Engineering (CASE) emerged in the late 1980s
 110 as the first systematic attempt to automate the software development lifecycle. CASE technology encompasses a collection of automated tools and methods that assist software engineering across its phases (Sommerville, 1996; Fuggetta, 1993), including editing tools, programming tools, code generators, verification and validation tools, configuration management systems, and project management tools (Fuggetta, 1993). Fuggetta (1993) classified
 115 CASE products into three categories: individual tools supporting specific tasks, workbenches integrating multiple tools for specific activities, and environments providing holistic toolkits supporting the entire software process.

The CASE market demonstrated significant commercial interest, generating revenues that grew to \$12.11 billion by 1995 (Sanchis et al., 2020). However, adoption rates were dramatically disappointing. Iivari (1996) found that 73.5% of companies never adopted CASE tools, concluding that the
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productivity and quality benefits were “overrated” and that complexity was a “significant, negatively related predictor of CASE effectiveness.” Lundell and Lings (2004) identified that the conceptual gap between tool designers and end users was a primary barrier. Huff (1992) demonstrated that training costs often exceeded the original tool purchase price. Orlikowski (1993) found that benefits did not accrue homogeneously, leading to internal resistance. These failures offer critical lessons: technological sophistication alone does not guarantee adoption, and unrealistic expectations can doom even promising paradigms. These failures bear instructive parallels to the present vibe-coding moment: both paradigms emerged amid economic pressure to compress development cycles, both promised to compress the gap between specification and implementation, and both depended on tool maturity outpacing practice maturity. We return to this comparison in Section 5.

2.1.2. Rapid Application Development, IDEs, and Web Frameworks (1990s–Early 2010s)

Following CASE’s mainstream rejection, software development automation entered a more pragmatic phase characterised by smaller-scale, developer-centric productivity gains rather than wholesale lifecycle automation. This roughly twenty-year interregnum produced four overlapping waves of innovation that progressively lowered the barrier to building functional software while preserving developer authorship.

Rapid Application Development (RAD) tools. Building on Martin’s (1991) RAD methodology, products including Microsoft Visual Basic (1991), Borland Delphi (1995), and Sybase PowerBuilder (1991) introduced visual form designers, event-driven programming models, and pre-packaged user-interface components. RAD tools democratised client–server application building among intermediate developers and presaged the citizen-developer ethos later articulated by low-code vendors (Martin, 1991; Beynon-Davies et al., 1999). Crucially, RAD succeeded commercially where CASE had failed because it augmented rather than displaced developer agency, generated code in mainstream languages, and exhibited substantially shorter learning curves (Beynon-Davies et al., 1999).

Integrated Development Environments (IDEs). The late 1990s and 2000s saw the maturation of the IDE as a primary productivity locus. Microsoft Visual Studio (1997) and the Eclipse platform (2001), followed by IntelliJ IDEA (2001) and JetBrains’ family of language-specific environments, embedded refactoring, debugging, version control, intelligent code comple-

tion, and static analysis directly into the developer workflow (Murphy et al., 2006; Storey et al., 2005). Murphy et al. (2006) demonstrated empirically that mature IDE tooling produced measurable productivity gains, contradicting the CASE-era assumption that productivity required full-lifecycle automation. The IDE paradigm established the foundation on which AI-assisted development tools—Cursor, GitHub Copilot, Claude Code—would later be built.

Web application frameworks. A parallel revolution unfolded in web development. Ruby on Rails (2004) introduced “convention over configuration” as an explicit philosophy of reducing developer choices to accelerate productive output. Django (2005), ASP.NET (2002), Spring (2003), and Laravel (2011) operationalised similar opinionated abstractions, automating database migrations, request routing, authentication, and templating. Empirical comparisons reported substantial reductions in lines of code and corresponding development-time savings relative to unstructured implementations (Plekhanova, 2009). These frameworks accustomed enterprises to the trade-off central to all subsequent automation paradigms: configurational productivity in exchange for the abstraction’s particular constraints.

Model-Driven Development (MDD). The Object Management Group’s Model-Driven Architecture initiative (Object Management Group, 2003) attempted a more ambitious automation strategy reminiscent of CASE—defining platform-independent models from which platform-specific code could be generated automatically (Schmidt, 2006). Hutchinson et al. (2011) surveyed industrial adoption of MDD and found benefits in productivity and maintainability for organisations that committed to the methodology, but limited diffusion overall, citing tooling immaturity, organisational change costs, and round-trip engineering difficulties. MDD’s mixed reception foreshadowed low-code’s architectural dilemmas (Cabot, 2020) and offers a cautionary lesson for vibe coding: code-generation paradigms demand sustained organisational investment to realise their promise.

Agile methodologies and DevOps. Concurrent with these tooling advances, the Agile Manifesto (Beck et al., 2001) and the subsequent DevOps movement (Humble and Farley, 2010; Forsgren et al., 2018) reshaped *how* software was developed. By replacing waterfall with iterative delivery and embedding continuous integration, automated testing, and infrastructure-as-code, agile and DevOps amplified the impact of every preceding tooling improvement: shorter development cycles became economically meaningful only when paired with shorter release cycles. Together, agile, DevOps, and

the IDE/framework stack established the operational substrate on which low-code platforms (2014) and vibe coding (2025) could plausibly promise enterprise-grade productivity gains.

This intermediate phase is critical for our analysis. It demonstrates that successful development-automation paradigms tend to share three properties: (a) they augment rather than replace developer agency; (b) they produce mainstream code or close substitutes; and (c) they are paired with complementary process innovations. Vibe coding’s eventual sustainable adoption may hinge on its capacity to satisfy these same conditions.

2.1.3. Low-Code/No-Code Platforms (2014–Present)

The term “low-code” was first coined by Forrester Research in 2014 (Richardson and Rymer, 2014) to describe platforms that enable application development through visual interfaces, pre-built components, and minimal hand-coding. Unlike CASE tools, which primarily assisted professional developers, low-code platforms explicitly targeted a broader user base including “citizen developers”—business professionals without formal programming training (OutSystems, 2019). Richardson and Rymer (2016) identified low-code platforms as providing useful solutions for automating and accelerating application delivery.

Sanchis et al. (2020) systematically documented the benefits of low-code platforms: (a) privacy—applications can be developed internally without outsourcing; (b) rapidity—Forrester surveys showed 5–10× acceleration in development speed (Richardson and Rymer, 2016); (c) cost reduction—shorter development cycles directly reduce project costs; (d) complexity reduction—apps are configured rather than coded from scratch; (e) easy maintenance—minimal code means minimal maintenance burden; (f) involvement of business profiles—44% of low-code platform users are business users collaborating with IT (OutSystems, 2019); and (g) minimization of unstable requirements—rapid iteration enables faster validation of ideas. The State of Application Development report (OutSystems, 2019) found that 66% of respondents cited “accelerate digital transformation” as their primary motivation for adopting low-code platforms.

However, significant limitations persist. Tisi et al. (2019) identified three critical constraints: scalability (low-code platforms are primarily suited for small applications, not enterprise-grade systems), fragmentation (different vendors implement different paradigms with limited interoperability), and restriction to software-only systems (manufacturing and IoT use cases requir-

ing hardware integration are poorly served). By 2025, the low-code market had grown substantially—Forrester projected total spending of \$21.2 billion (Rymer and Koplowitz, 2019)—yet these fundamental limitations continued to constrain its role in enterprise digital transformation. Recent empirical work has begun to qualify these limitations: Bock and Frank (2021) document successful enterprise-scale low-code deployments where governance was embedded ex ante, and Prinz et al. (2021) identify “low-code platform engineering” as an emergent organizational discipline mediating between citizen and professional developers. These findings suggest that the central question for any successor paradigm—including vibe coding—is whether governance can mature in step with diffusion.

2.1.4. *Vibe Coding (2025–Present)*

On February 2, 2025, Andrej Karpathy introduced “vibe coding” as a practice where developers describe desired functionality in natural language, allow LLMs to generate complete code, and “accept all” without necessarily comprehending every line of output (Karpathy, 2025). The defining characteristics of vibe coding, as synthesized from subsequent literature, include: (a) natural language as the primary development interface (Fawzy et al., 2025; Various Authors, 2025b); (b) AI-generated code as the primary output in standard programming languages rather than platform-specific configurations (Various Authors, 2025c); (c) an “accept all” mentality (Karpathy, 2025; Fawzy et al., 2025); (d) iterative error resolution through AI (Various Authors, 2025b); and (e) code that may exceed developer comprehension (Karpathy, 2025; Various Authors, 2025e). These characteristics jointly distinguish vibe coding from earlier AI-assisted development practices (e.g., GitHub Copilot circa 2021–2024), in which AI suggestions remained subordinate to a developer review loop (Ziegler et al., 2022; Bird et al., 2023), by inverting the locus of authorship: the developer becomes a curator of AI output rather than its author.

The vibe coding ecosystem rapidly matured around several key platforms: Cursor (an AI-native code editor), Replit (cloud-based IDE with AI), Claude Code (Anthropic’s command-line AI tool), GitHub Copilot (comprehensive AI coding assistant), and newer platforms including v0.dev, Bolt, and Lovable that specifically target non-developers (Fawzy et al., 2025; Various Authors, 2025c).

Willison (2025) introduced an important distinction between “vibe coding” and what he termed “vibe engineering”—the practice of using AI coding

tools while maintaining full accountability, conducting code reviews, writing tests, and ensuring comprehension of the output. This distinction highlights a spectrum of AI-assisted development practices with significant implications for code quality, security, and organizational resilience.

Current adoption statistics underscore the paradigm’s momentum. A 2025 Stack Overflow survey found that 92% of U.S. developers use AI coding tools daily (Stack Overflow, 2025). Gartner predicted that by 2028, 75% of enterprise software engineers would use AI coding assistants (Gartner, 2025). McKinsey & Company (2025) indicated that 88% of organizations report regular AI use across functions. Yet caution is warranted: Boston Consulting Group (2025) found that 60% of organizations generate no material value from their AI investments, suggesting that adoption does not automatically translate to organizational benefit. This adoption-impact gap parallels findings from prior digital-transformation research showing that 60–70% of large-scale technology deployments fail to translate into measurable business value (Tabrizi et al., 2019; Saldanha, 2019), suggesting that AI coding’s fate may depend less on its intrinsic capabilities than on the surrounding organizational architecture of governance, capability building, and value-capture mechanisms.

2.2. Digital Transformation and Organizational Resilience

2.2.1. Enterprise and Digital Resilience

Enterprise resilience has been defined as the capacity to prevent, anticipate, change, and adapt to changing environments (Sanchis and Poler, 2019). In the digital context, Heeks and Ospina (2019) defined digital resilience as “the capabilities developed through digital technologies to absorb shocks, adapt, and transform in the face of adverse change.” This conceptualization encompasses three progressive stages: anticipation (detecting threats and opportunities early), coping (maintaining operations during disruption), and adaptation (learning and transforming in response to change) (Heeks and Ospina, 2019; Holling, 1973).

The relationship between digital transformation and resilience is increasingly well-documented. Browder (2024) characterized digital transformation as “upgrading adaptation”—an investment that enhances an organization’s capacity to respond to future disruptions. Li and Wang (2024) demonstrated empirically that digital transformation positively influences enterprise resilience. Breidbach et al. (2024) introduced the concept of “orchestrating

digital resilience,” emphasizing the active role of organizational decision-making. McKinsey research indicates that agile organizations make decisions
310 50% faster than their peers (McKinsey & Company, 2023), directly linking development speed to organizational resilience. These findings align with broader theorising on *ambidextrous* digital resilience (Boh et al., 2022), in which organizations must simultaneously exploit current digital capabilities and explore emerging ones. Vibe coding sits squarely within this ambidexterity tension: it lowers the marginal cost of exploration but, if ungoverned,
315 may erode the foundations of exploitation.

2.2.2. Rapid Development Capability as a Resilience Enabler

A critical but under-theorized link exists between software development speed and organizational resilience. When organizations can build, test, and
320 deploy digital solutions rapidly, they gain the ability to: (a) detect market opportunities and threats through rapid prototyping and MVP testing; (b) respond to disruptions by building digital solutions in days rather than months; and (c) continuously adapt their digital infrastructure as conditions change. This positions development capability—specifically, the speed and
325 accessibility of software creation—as a direct enabler of digital resilience. Both low-code platforms and vibe coding dramatically compress development timelines, but through fundamentally different mechanisms (Sanchis et al., 2020; Fawzy et al., 2025).

From a strategic perspective, enterprise resilience has been framed as a
330 function of *anticipatory capacity*—the organizational ability to construct and revise plausible futures (Rohrbeck and Kum, 2018). Vibe coding’s central question for digital business is whether it strengthens or atrophies this capacity. By dramatically lowering the cost of building exploratory prototypes, it may expand the futures organizations can plausibly explore; by potentially
335 eroding deep technical understanding, it may simultaneously narrow the realistic horizon of what those organizations can build and operate. This tension is what our integrated theoretical framework, presented next, is designed to surface.

2.3. Theoretical Lens

340 This study employs an integrated multi-theory framework combining Dynamic Capabilities Theory, the TOE framework, and DOI Theory. This integration is necessary because no single theory adequately captures both the

adoption dynamics and the organizational impact of vibe coding on digital resilience.

345 2.3.1. *Dynamic Capabilities Theory*

Teece et al. (1997) defined dynamic capabilities as “the firm’s ability to integrate, build, and reconfigure internal and external competences to address rapidly changing environments.” Warner and Wäger (2019) extended this framework to the digital transformation context, identifying nine digitally
350 based microfoundations underpinning three core capability clusters:

- **Sensing:** The capacity to detect opportunities and threats. Vibe coding enhances sensing by enabling rapid prototyping (hours instead of months), allowing organizations to test market hypotheses at dramatically lower cost and higher speed (Warner and Wäger, 2019; Teece,
355 2018).
- **Seizing:** The capacity to mobilize resources to capture value from opportunities. Vibe coding compresses time-to-market from months to days or hours, enabling organizations to capture value before competitors (Teece et al., 1997; Warner and Wäger, 2019).
- 360 • **Transforming:** The capacity to reconfigure organizational assets and structures. Vibe coding empowers citizen developers, fundamentally reconfiguring the boundary between IT and business functions (Warner and Wäger, 2019; Lethbridge, 2024).

2.3.2. *Technology-Organization-Environment (TOE) Framework*

365 Tornatzky and Fleischer’s (1990) TOE framework structures the analysis of technology adoption factors across three contexts: *technology context* (LLM capability maturity, tool accessibility, code quality trade-offs, integration capabilities); *organizational context* (existing digital maturity, AI culture and readiness, management support, governance structures); and *envi-*
370 *ronmental context* (competitive pressure, industry-specific regulations, talent market dynamics, vendor ecosystem evolution).

2.3.3. *Diffusion of Innovation (DOI) Theory*

Rogers’s (2003) DOI theory identifies five innovation characteristics that influence adoption rates. Vibe coding exhibits: exceptional *relative advantage* (60–80% faster prototyping, substantial cost reduction); high *compati-*
375 *ty*

380 *bility* with human communication patterns but low compatibility with engineering practices; near-zero *complexity* (natural language interface); high *trialability* (free/low-cost tools); and high *observability* (working prototypes in hours). Based on current adoption patterns, vibe coding appears to be transitioning from the “early adopters” to the “early majority” stage of Rogers’ diffusion curve (Rogers, 2003).

2.3.4. Integrated Conceptual Model

Our integrated model posits that vibe coding adoption (driven by TOE and DOI factors) enhances organizational dynamic capabilities (sensing, seizing, transforming), which in turn produce digital resilience outcomes (anticipation, coping, adaptation). This relationship is moderated by industry sector, organizational size, existing digital maturity, and governance maturity. Critically, the model also identifies sustainability tensions that may undermine resilience outcomes. Fig. 1 presents this integrated conceptual model.

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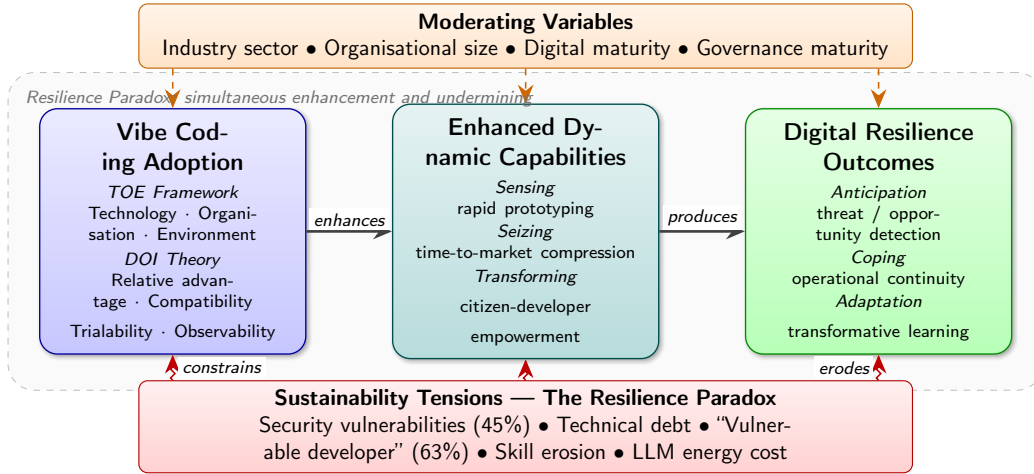


Figure 1: Integrated conceptual model: vibe coding adoption (TOE+DOI factors) enhances organisational dynamic capabilities, which produce digital resilience outcomes. The horizontal black flow is the primary causal path. Dashed orange arrows indicate moderating influences from contextual variables. Zigzag red arrows indicate sustainability tensions that simultaneously undermine each stage of the flow—the resilience paradox in which vibe coding both strengthens and weakens organisational digital resilience.

The justification for this multi-theory approach is fourfold and follows re-

cent calls for *theoretical pluralism* in digital-transformation research (Hanelt et al., 2021). First, TOE and DOI together provide complementary explanations for adoption variance: TOE specifies the contextual factors shaping adoption decisions, while DOI specifies the innovation properties that determine diffusion velocity. Neither alone explains why an innovation with vibe coding’s exceptional DOI profile (high relative advantage, near-zero complexity) faces uneven TOE-mediated adoption across sectors. Second, Dynamic Capabilities Theory translates adoption into organizational impact, opening the analytical space between technology use and resilience outcomes that single-theory adoption studies typically leave underspecified (Warner and Wäger, 2019; Teece, 2018). Third, the integration enables identification of *paradoxes*—configurations in which the same factor produces opposing effects on different resilience dimensions—that single-theory approaches systematically miss. Fourth, theoretical pluralism is methodologically aligned with directed content analysis (Hsieh and Shannon, 2005), which uses theoretical frameworks as initial coding scaffolds while remaining open to inductive extension.

3. Research Methodology

3.1. Research Design and Justification

This study employs directed content analysis following Hsieh and Shannon (2005), supplemented by Mayring’s (2014) systematic procedures and Elo and Kyngäs’s (2008) preparation-organizing-reporting structure. Directed content analysis begins with existing theory as initial coding guidance—in our case, the integrated Dynamic Capabilities, TOE, and DOI framework—while remaining open to new categories that emerge inductively from the data (Hsieh and Shannon, 2005). This approach is particularly appropriate for three reasons. First, vibe coding is an emerging phenomenon with fragmented discourse across heterogeneous sources; content analysis enables systematic synthesis. Second, the directed approach leverages established theoretical frameworks while accommodating novel dynamics. Third, multi-source design enables triangulation that strengthens validity. Sample-size adequacy was assessed via *thematic saturation* (Saunders et al., 2018): by source 80, no new meta-categories or sub-categories emerged across two consecutive review batches of 10 sources each, indicating that the n=97 corpus was sufficient for the research questions.

3.2. Data Collection

3.2.1. Search Strategy

Data collection employed a systematic multi-source triangulation strategy across four source types, as summarized in Table 1.

Table 1: Data source types, search strategies, and corpus distribution.

Source Type	Scope	Search Strategy	Retrieved	Included
Academic literature	Peer-reviewed papers on vibe coding, AI-assisted development, low-code + DT + resilience	Scopus, Web of Science, IEEE Xplore, ACM DL, AISEL; PRISMA 2020	127	34
Grey literature	Industry reports, whitepapers, enterprise case studies	Gartner, Forrester, McKinsey, BCG, Atos, HCLTech, Genpact; Garousi et al. (2019) guidelines + AACODS	89	31
Developer community	Practitioner blogs, forums, social media	Stack Overflow, GitHub, Reddit, X/Twitter, DEV Community, Medium	112	22
Platform documentation	Official docs, enterprise case studies	Cursor, Replit, Claude Code, Copilot, v0.dev, Bolt, Lovable + OutSystems, Mendix, Power Apps	38	10
Total			366	97

Search keywords were organized into four clusters: (a) primary terms: “vibe coding,” “AI-assisted coding,” “AI-assisted development,” “LLM coding,” “vibe engineering”; (b) combined with: “digital transformation,” “digital resilience,” “enterprise resilience,” “organizational resilience,” “citizen developer”; (c) comparative terms: “low-code,” “no-code,” “CASE tools”; (d) temporal scope: February 2025 to February 2026 for vibe coding; 2014–2026 for low-code comparative sources.

3.2.2. Inclusion and Exclusion Criteria

Table 2 presents the inclusion and exclusion criteria applied across all source types.

3.2.3. Quality Assessment

Academic sources were assessed through the peer-review process as a baseline quality indicator. Grey literature was assessed using the AACODS check-

Table 2: Inclusion and exclusion criteria.

Criteria	Include	Exclude
Language	English	Non-English
Timeframe	Feb 2025–Feb 2026 (vibe coding); 2014–2026 (low-code/DT)	Before 2014 for comparative sources
Focus	Organizational/business impact on DT/resilience	Purely technical benchmarks without organizational context
Quality	Peer-reviewed OR grey lit $\geq$ 10/15 on AACODS	Marketing materials; sources $<$ 10/15 quality
Relevance	Adoption, impact, benefits, bar- riers at organizational level	Solely individual developer pro- ductivity metrics

list (Tyndall, 2010) (Authority, Accuracy, Coverage, Objectivity, Date, Sig-
445 nificance), with a threshold of 10/15 for inclusion, following Garousi et al.’s
(2019) guidelines. For instance, a Forrester report citing original survey
methodology with $n > 500$ respondents and clear sampling frames scored
13/15 and was retained; a generic vendor whitepaper with promotional lan-
guage and undisclosed methods scored 6/15 and was excluded. Developer
450 community sources were evaluated for substantive analytical content, author
credibility, and engagement metrics. The full screening process from initial
retrieval ($n = 366$) to final inclusion ($n = 97$) follows the PRISMA 2020
reporting guidelines (Page et al., 2021).

3.2.4. Corpus Description

455 The final corpus comprises 97 sources: 34 academic papers (35.1%), 31
grey literature sources (32.0%), 22 developer community sources (22.7%),
and 10 platform documentation sources (10.3%). Geographically, sources
originate from North America (48.5%), Europe (27.8%), Asia-Pacific (16.5%),
and other regions (7.2%).

3.3. Data Analysis Procedure

3.3.1. Hierarchical Coding Framework

460 Following the directed content analysis approach (Hsieh and Shannon,
2005), we developed a hierarchical coding framework with deductive cate-
gories derived from our theoretical framework and inductive subcodes that
465 emerged during analysis. Table 3 presents the framework.

Table 3: Hierarchical coding framework.

Meta-Category (Deductive)	Sub-Categories (Deductive)
1. Paradigm Characteristics	Natural language interface; AI-generated code; acceptance without review; iterative error resolution; code exceeding comprehension
2. Adoption Drivers	Speed/efficiency; cost reduction; accessibility/democratization; skills gap mitigation; competitive pressure; innovation velocity
3. Adoption Barriers	Security vulnerabilities; technical debt; governance gaps; quality concerns; skill erosion; vendor/AI dependency; regulatory compliance
4. Dynamic Capability Enhancement	Rapid prototyping (sensing); time-to-market compression (seizing); citizen developer empowerment (transforming); organizational boundary reconfiguration
5. DT Enablement	Cross-functional development; legacy modernization; process automation; customer-facing innovation; internal tool building
6. Resilience Outcomes	Operational continuity; adaptive capacity; innovation velocity; knowledge retention/loss; workforce resilience
7. Sustainability Tensions	Resource efficiency vs. LLM energy costs; democratization vs. quality erosion; speed vs. technical debt; agility vs. governance

3.3.2. Coding Process

The unit of analysis was the meaningful text segment at the paragraph or quote level. Analysis proceeded in three phases: (1) preparation—corpus compilation, quality assessment, and familiarization; (2) organizing—deductive coding using the a priori framework, followed by inductive coding; and (3) reporting—synthesis, frequency analysis, and thematic integration (Elo and Kyngäs, 2008). All coding was conducted using NVivo 14. Analytical memos were maintained throughout coding to capture interpretive decisions, paradox observations, and theoretical insights, following Saldaña (2021); theoretical saturation was monitored prospectively at the meta-category level.

3.4. Quality Assurance

Intercoder reliability. Two researchers independently coded 20% of the corpus (20 sources stratified across all four source types). Krippendorff’s α was computed at the meta-category level across all coded sources $\times$ 7 categories. Initial Krippendorff’s Alpha was 0.74. Following discussion, refinement of coding definitions, and a second round, Alpha reached 0.83, exceeding the 0.80 threshold recommended by Krippendorff (2019). Disagreements were resolved through structured negotiation following Campbell et al. (2013).

Trustworthiness. Following Lincoln and Guba (1985), we addressed: credibility through triangulation and member checking with three industry practitioners; transferability through thick description; dependability through a detailed audit trail; and confirmability through a reflexive journal.

Pilot coding. The framework was pilot-tested on 10% of the corpus (10 sources), resulting in refinement of three sub-category definitions and addition of two inductive subcodes.

Reflexivity. Following Nicmanis (2024), we acknowledge our positionality as information systems researchers with prior experience in digital transformation and technology adoption.

4. Findings

4.1. Descriptive Overview of the Corpus

The 97-source corpus reveals a discourse landscape dominated by practitioner and industry voices. Academic literature constitutes only 35.1% of the corpus, reflecting the recency of the phenomenon. Within academic sources, 73.5% are preprints or conference papers rather than journal articles.

The most frequently coded meta-categories were Adoption Drivers (coded in 81.4% of sources), Paradigm Characteristics (76.3%), and Adoption Barriers (74.2%). Dynamic Capability Enhancement (52.6%) and Digital Resilience Outcomes (41.2%) were less frequently coded, confirming that organizational-level impact analysis remains underdeveloped.

Temporal analysis reveals exponential growth: 8 sources in February–April 2025 (the “naming period”), 23 in May–July (“awareness expansion”), 31 in August–October (“critical analysis”), and 35 in November 2025–February 2026 (“maturation”), suggesting rapid evolution from excitement toward nuanced assessment.

4.2. The Vibe Coding Paradigm: Characterization and Evolution (RQ1)

4.2.1. Discourse Across Source Types

A significant divergence exists between academic and practitioner framings. Academic sources predominantly frame vibe coding as a productivity tool with associated risks, employing quantitative metrics (Fawzy et al., 2025; Various Authors, 2025b,e). Practitioner and industry sources frame it as a paradigm shift with organizational implications, using language of “democratization,” “disruption,” and “transformation” (Shmargad, 2025; McKinsey & Company, 2025; Atos, 2025). Developer community sources exhibit the widest range, from enthusiastic advocacy to deep skepticism (Various Authors, 2025a,d).

4.2.2. Positioning on the CASE → Low-Code → Vibe Coding Continuum

Our analysis identifies vibe coding as the fourth major paradigm in software development automation, following CASE tools, the RAD/IDE/web-framework era, and low-code platforms. Table 4 presents a systematic comparison, extending Sanchis et al.’s (2020) benchmarking framework (their Table 5) to include the intermediate era and vibe coding.

This comparative analysis reveals that each paradigm addressed specific limitations of its predecessor while introducing new challenges. Vibe coding addresses accessibility (natural language interface) and eliminates vendor lock-in (standard code output) but introduces novel risks around code quality, security, and the decoupling of code creation from code comprehension.

Table 4: Comparative analysis of development automation paradigms (extending Sanchis et al. (2020)).

Feature	CASE (1980s–90s)	Tools	RAD / Web frameworks (1990s–early 2010s)	IDEs Frame- works	Low-Code (2014–present)	Vibe (2025–present)	Coding
Development interface	Visual modeling (UML), structured design		Visual designers + manual code; rich editors; convention-driven scaffolding		Visual drag-and-drop, pre-built modules	Natural language prompts to LLMs	
Code generation	Automated models (often partial)	from (often)	Manual + scaffolded boilerplate		Platform-specific configs	Standard code (HTML, CSS, JS, Python)	
Code ownership	Developer-owned, tool-dependent formats		Developer-owned, mainstream languages		Vendor-locked / platform-dependent	Full ownership, deploy anywhere	
Target user	Professional developers		Professional + power-user developers		Citizen developers (some training)	Anyone with domain knowledge	
Learning curve	High (specialized training)		Moderate (language + framework)		Moderate (platform-specific)	Near-zero (natural language)	
Security governance	Developer-managed		Developer-managed; framework defaults		Platform-embedded controls	Unstructured; 45% vulnerabilities	
Enterprise scalability	Designed for enterprise; poor adoption		Strong (production-proven)		Proven for departmental apps	Uncertain; complexity ceiling	
Development speed	Modest improvement		2–3× faster than greenfield		5–10× faster	60–80% faster; prototypes in hours	
Cost model	High license + training		License or open source		Platform subscription	AI API costs (often lower)	
QA practices	Integrated verification tools		Test-driven culture (xUnit)		Built-in testing	Often skipped (36% skip QA)	
Vendor lock-in	Moderate		Low to moderate		High (platform dependency)	Low (standard code output)	
Maintenance burden	Moderate		Moderate		Low (platform-managed)	High (AI code harder to maintain)	
Governance maturity	Moderate		Mature; codified by DevOps		Established (compliance features)	Nascent (no built-in governance)	
Adoption rate	26.5% peak		Mainstream		Growing; \$21.2B market	87% Fortune 500 in 12 months	
Key failure mode	Unrealistic expectations, complexity		Framework churn, abstraction leakage, MDD round-tripping		Vendor lock-in, scalability	Security, tech debt, skill erosion	

4.3. Vibe Coding and Digital Resilience: Enablers and Threats (RQ2)

4.3.1. Resilience Enablers Mapped to Dynamic Capabilities

535 *Sensing enhancement..* Forty-seven sources (48.5%) identify vibe coding’s rapid prototyping capability as a significant enhancer of organizational sensing. The ability to build functional MVPs in hours rather than months allows organizations to test market hypotheses at dramatically reduced cost. One industry report describes a financial services firm that used vibe coding to build and deploy seven prototype customer-facing applications in a single sprint (Atos, 2025). Illustratively, one practitioner-coded source characterised enterprise rapid-prototyping practice as “the difference between a strategy meeting that produces a slide deck and a strategy meeting that produces a working app” (McKinsey & Company, 2025)—a framing that recurs across 14 of the 47 sensing-coded sources, suggesting a coherent practitioner narrative around speed-of-hypothesis as a sensing capability. This “hypothesis velocity” directly enhances the sensing dimension of dynamic capabilities as theorized by Warner and Wäger (2019).

550 *Seizing acceleration..* Fifty-three sources (54.6%) identify compressed development cycles as the primary seizing enhancement. The reduction of time-to-market from months to days or hours enables organizations to capture value before competitors. Cost reduction amplifies seizing by lowering the threshold for pursuing opportunities—McKinsey & Company (2025) estimates that organizations using AI-assisted development reduce costs by 30–50% for appropriate use cases.

560 *Transforming capability..* Thirty-nine sources (40.2%) identify citizen developer empowerment as a transformative capability. When non-technical staff can build functional applications through natural language, the boundary between IT and business functions fundamentally shifts. This manifests as: business units building their own tools; domain experts creating specialized applications capturing tacit knowledge; distributed innovation; and a broader base of digitally capable employees creating resilience to talent loss (Lethbridge, 2024; HCLTech, 2025).

4.3.2. Threats to Resilience

565 *Security vulnerabilities..* The most frequently coded threat (60.8% of sources). Veracode’s (2025) report found that 45% of AI-generated code introduces security vulnerabilities. CodeRabbit (2025) reported $2.74\times$ higher security

issues in AI co-authored code. The threat is compounded by the “accept all” mentality characteristic of vibe coding—vulnerabilities propagate into production undetected.

The “vulnerable developer” phenomenon.. Thirty-one sources (32.0%) discuss what Fawzy et al. (2025) term the “vulnerable developer”—a practitioner who can build applications but cannot debug, maintain, secure, or understand the code. An estimated 63% of active vibe coding practitioners are non-developers lacking maintenance skills. The 63% figure derives from triangulating Fawzy et al. (2025)’s grey-literature review, the GitHub Octoverse 2025 report (GitHub, 2025), and developer-community thread analysis (Various Authors, 2025a); we treat it as a discourse-derived plausibility benchmark requiring future quantitative validation. This directly undermines the coping dimension of resilience.

Technical debt accumulation.. Forty-four sources (45.4%) identify technical debt as a significant concern. The “flow-debt trade-off” (Various Authors, 2025e) captures this dynamic: frictionless code generation encourages rapid production, but the code often exhibits architectural inconsistencies, redundant logic, and violations of best practices. Since maintenance represents over 80% of total software lifecycle cost (Pressman and Maxim, 2020), fast-to-build but expensive-to-maintain code does not necessarily reduce total cost of ownership.

Skill erosion and knowledge concentration risk.. Twenty-six sources (26.8%) raise concerns about long-term skill erosion. When developers routinely accept AI-generated code without engaging in design and debugging, core engineering skills may atrophy, creating fragile AI dependency (Fawzy et al., 2025; Various Authors, 2025d).

4.3.3. Comparative Resilience Analysis

Table 5 presents a comparative analysis of resilience factors across low-code and vibe coding paradigms.

This comparison reveals that vibe coding amplifies both the positive (speed, cost, democratization) and negative (security, maintenance, knowledge fragility) resilience dimensions. The key differentiator is governance: low-code embeds governance within platforms, while vibe coding places the entire burden on the organization.

Table 5: Comparative resilience analysis: Low-code vs. vibe coding.

Resilience Factor	Low-Code	Vibe Coding	Assessment
Development speed	5–10× faster	60–80% faster; hours	VC significantly amplifies
Cost of experimentation	Moderate (subscription)	Low (API costs)	VC lowers barrier further
Citizen developer enablement	Partial (training needed)	Extensive (NL interface)	VC dramatically extends
Security governance	Platform-embedded	No built-in governance	LC significantly stronger
Code maintainability	Platform-managed updates	Developer-managed; poor docs	LC significantly stronger
Vendor lock-in risk	High (platform dependency)	Low (standard code)	VC significantly stronger
Knowledge retention	Moderate	Low (code > comprehension)	LC moderately stronger
Governance maturity	Established	Nascent	LC significantly stronger

4.4. Adoption Factors and Sustainability Implications (RQ3)

4.4.1. TOE Analysis of Adoption Factors

Technology factors. The most cited driver is LLM capability trajectory (67.0%), followed by tool ecosystem maturity (54.6%). The primary barrier is integration challenges (41.2%)—connecting AI-generated code with existing enterprise systems requires expertise contradicting the accessibility promise (Fawzy et al., 2025; Various Authors, 2025b).

Organizational factors. Existing digital maturity emerges as the strongest organizational predictor (52.6%). Organizations with established development workflows can integrate vibe-coded contributions more safely. Governance readiness (38.1%) is critical but often absent (HCLTech, 2025; Genpact, 2025).

Environmental factors. Competitive pressure is the most cited driver (61.9%): with 87% of Fortune 500 adopting (Shmargad, 2025), non-adopters face strategic anxiety. The talent shortage (49.5%) drives adoption as a substitute for scarce human resources. Regulatory uncertainty (35.1%) acts as both barrier and driver.

4.4.2. DOI Analysis

Vibe coding exhibits an unusual combination of innovation characteristics: exceptionally high relative advantage, trialability, observability, and low complexity, but poor compatibility with existing engineering practices. This creates what we term a “*quality chasm*”—rapid diffusion for prototyping and non-critical applications, with slower adoption for production and mission-critical systems. This bifurcated diffusion pattern resembles the two-speed adoption phenomenon documented by Christensen et al. (2015) for disruptive technologies, in which the technology rapidly captures low-end and adjacent-market segments while encountering durable resistance in mainstream high-stakes domains. Whether vibe coding crosses this chasm for enterprise applications (Moore, 2014) is conditional on what we term “*governance-equivalence*”: the moment at which vibe coding’s risk profile becomes commensurable with established enterprise software risk profiles via maturing tooling, frameworks, and accountability structures.

4.4.3. Sustainability Tensions

Four primary tensions emerge:

1. **Speed vs. quality** (63.9%): 62% are motivated by speed but 68% perceive outputs as “fast but flawed” (Fawzy et al., 2025). The speed advantage derives partly from skipping quality assurance steps.
- 640 2. **Democratization vs. governance** (38.1%): When anyone can build and deploy applications, shadow IT risk increases. Sanchis et al. (2020) identified this tension in low-code; vibe coding amplifies it.
3. **Innovation velocity vs. technical debt** (35.1%): Rapid creation of hard-to-maintain code may generate negative long-term value (Various Authors, 2025e; Pressman and Maxim, 2020).
- 645 4. **Resource efficiency vs. environmental cost** (16.5%): LLM inference requires significant energy; generating code through AI shifts environmental burden from human labor to data center computation (Strubell et al., 2019).

5. Discussion

650 5.1. Theoretical Contributions

5.1.1. The Resilience Paradox

The central theoretical contribution is the identification of the “*resilience paradox*” of vibe coding. Through Dynamic Capabilities Theory, we find that vibe coding simultaneously strengthens and weakens organizational digital
655 resilience. It strengthens sensing (rapid prototyping), seizing (compressed time-to-market), and transforming (citizen developer empowerment). Yet it introduces fragilities that undermine coping (security vulnerabilities, un-maintainable code) and may compromise adaptation (skill erosion, knowledge concentration risk).

660 This extends Warner and Wäger’s (2019) framework. Their nine micro-foundations assume digital capabilities are built by competent actors who understand the tools and systems they create. Vibe coding challenges this assumption. We propose an extension introducing a “*capability quality*” dimension alongside capability speed and scope—recognizing that faster capability building is only beneficial if the outputs are reliable, secure, and
665 maintainable.

5.1.2. TOE Refinement for AI-Assisted Development

Our findings refine TOE in three ways. First, the technology dimension must account for AI model capability as a *dynamically evolving* factor—
670 unlike traditional technologies with stable feature sets, LLM capabilities change with each model generation. Second, the organizational dimension must incorporate “AI readiness” as distinct from traditional IT readiness. Third, the environmental dimension is characterized by unusually intense competitive pressure and FOMO dynamics that may drive premature adop-
675 tion.

5.1.3. DOI Insights and the Quality Chasm

Our DOI analysis reveals that vibe coding’s innovation attributes predict extremely rapid diffusion. However, the compatibility gap creates a “quality chasm” that may segment the market into two adoption trajectories:
680 rapid adoption for non-critical applications and slower adoption for enterprise-grade systems. This extends Rogers’s (2003) theory by identifying how a single innovation can follow different diffusion curves based on use-case criticality.

5.2. Practical Implications for Organizations

685 5.2.1. Governance Framework

We propose a tiered governance framework aligned with application criticality:

Tier 1—Exploration and prototyping (minimal governance). For internal prototypes and disposable experiments. No production deployment,
690 no sensitive data.

Tier 2—Internal tools and departmental applications (moderate governance). Requires automated security scanning, basic code review, integration testing, and documentation.

Tier 3—Customer-facing and business-critical (full governance).
695 Subject to comprehensive code review, security audit, performance testing, compliance verification, and maintenance planning.

5.2.2. The “Vibe Engineering” Middle Ground

Willison’s (2025) distinction between vibe coding and “vibe engineering” points toward a sustainable practice. Organizations may establish vibe en-
700 gineering as the standard for professional developers, reserving pure vibe coding for Tier 1 activities.

5.2.3. Hybrid Development Strategy

The optimal strategy deploys paradigms complementarily: *vibe coding* for prototyping and exploration; *low-code platforms* for production departmental applications requiring governance; and *traditional development* for mission-critical systems with stringent requirements.

5.2.4. Workforce Implications

Emerging roles include AI code reviewers, prompt engineers, and digital capability coordinators. Existing developers should be reskilled toward AI-augmented practices rather than replaced, as engineering expertise becomes more valuable when AI-generated code requires human oversight (Fawzy et al., 2025; Various Authors, 2025d).

5.3. Forecasting the Paradigm Shift

5.3.1. Hype Cycle Positioning

Applying the Gartner Hype Cycle, *vibe coding* in early 2026 appears to be approaching the “Peak of Inflated Expectations.” We anticipate a “Trough of Disillusionment” in 2026–2027 as organizations experience consequences of ungoverned *vibe-coded* applications at scale.

5.3.2. Historical Parallels with CASE Tools

Both CASE tools and *vibe coding* generated initial enthusiasm and promised democratization. However, critical differences suggest *vibe coding* may avoid CASE’s trajectory: dramatically lower learning curve, lower experimentation cost, immediately visible results, and an underlying technology on a steep improvement trajectory. The cautionary parallel is the quality gap—if AI-generated code remains fundamentally less reliable, organizations may confine *vibe coding* to narrow use cases.

5.3.3. Critical Factors for Sustainable Adoption

Five factors will determine mainstream adoption: (1) *security maturation*—AI tools must achieve substantially lower vulnerability rates; (2) *governance standardization*—industry-standard frameworks must emerge; (3) *maintenance tooling*—tools for maintaining AI-generated code must mature; (4) *regulatory clarity*—guidance for AI-generated code in regulated industries; and (5) *educational adaptation*—curricula must evolve for AI-augmented development.

735 5.3.4. *Scenario Forecasts: Three Trajectories to 2030*

Synthesising our findings across TOE, DOI, and Dynamic Capabilities, we project three plausible trajectories for vibe coding adoption through 2030. The actual trajectory is likely to be a weighted combination of these, with weights determined by the five critical factors enumerated above. We present
740 these as forecasting heuristics for organizational decision-makers, regulators, and educators, and identify observable signposts that distinguish them over the 2026–2028 window.

Scenario A — Stratified coexistence (most likely). Vibe coding establishes durable dominance in non-critical and exploratory contexts;
745 vibe engineering becomes the professional standard for departmental and customer-facing applications; traditional development persists for safety-critical systems. Governance frameworks mature into industry-specific norms similar to current cybersecurity-control standards. The 87% Fortune-500 adoption figure stabilises near 95%, but with high variance in maturity. Observable
750 signposts: industry bodies (e.g., NIST, ISO, sector regulators) publish vibe-coding governance guidance; major insurance carriers price AI-generated code risk; enterprise vendors release first-generation vibe-coding compliance tooling.

Scenario B — Quality breakthrough (plausible). Continued LLM
755 capability improvements (verifiable code generation, formal-method integration, model-level security alignment) close the quality gap. Vibe coding crosses the chasm into mission-critical applications; traditional development contracts to specialised domains (firmware, real-time systems, safety-critical kernels). Educational curricula reorganise around AI-augmented practice
760 as the default. Observable signposts: published AI-generated-code vulnerability rates fall below 10%; first regulated-industry deployments (banking, healthcare) of substantially AI-generated production systems are publicly documented.

Scenario C — Quality plateau and partial retreat (less likely).
765 A high-profile vibe-coding-related incident—a security breach, regulatory action, or systemic outage—triggers conservative organizational pullback. Governance becomes punitive. Vibe coding contracts to a narrow niche similar to that ultimately occupied by CASE tools, while traditional and low-code persist for serious work. Educational systems remain conservative. Observ-
770 able signposts: emergence of a high-impact vibe-coding incident receiving sustained media and regulatory attention; downward revision of AI-coding

adoption forecasts by major analyst firms; the appearance of “vibe coding excluded” clauses in enterprise procurement contracts.

These probabilities are subjective and reflect our reading of the current
775 evidence base; the scenarios are deliberately falsifiable against the signposts
above.

5.3.5. *Vision: The Hybrid Future*

The future is a layered coexistence of vibe coding, low-code, and tradi-
tional development, each serving different use cases along a criticality spec-
trum. Organizations developing governance frameworks, workforce capabili-
ties, and strategic clarity to deploy each paradigm appropriately will be best
780 positioned for digital resilience.

6. Conclusion, Limitations, and Future Research

6.1. *Summary of Contributions*

This study provides the first theoretically grounded, cross-industry con-
785 tent analysis of vibe coding’s impact on organizational digital transforma-
tion and resilience. Through directed content analysis of 97 sources analyzed
through Dynamic Capabilities, TOE, and DOI, we offer three principal con-
tributions.

Addressing **RQ1**, we characterize vibe coding as the fourth paradigm in
790 software development automation—following CASE tools, the RAD/IDE/web-
framework era, and low-code platforms—document academic–practitioner
discourse divergence, and provide a systematic comparative analysis extend-
ing Sanchis et al.’s (2020) framework.

Addressing **RQ2**, we identify and theorize the “*resilience paradox*”—vibe
795 coding simultaneously enhances sensing, seizing, and transforming capabil-
ities while introducing security vulnerabilities (45%), the “vulnerable devel-
oper” phenomenon (63%), technical debt, and skill erosion risk.

Addressing **RQ3**, we identify adoption factors through TOE analysis,
800 characterize the DOI “quality chasm,” and document four sustainability ten-
sions: speed vs. quality, democratization vs. governance, innovation velocity
vs. technical debt, and resource efficiency vs. environmental cost.

6.2. *Practical Takeaways*

For organizational leaders: (a) adopt a tiered governance framework
805 matching rigor to criticality; (b) establish vibe engineering as the professional

standard; (c) pursue hybrid development strategies; (d) invest in reskilling toward AI-augmented development; and (e) prepare for governance and maintenance challenges as vibe-coded applications mature.

6.3. Limitations

810 Several limitations must be acknowledged. First, the 12-month temporal scope captures a rapidly evolving phenomenon in its earliest stage. Second, English-only restriction excludes perspectives from non-English-speaking communities. Third, content analysis involves subjective interpretation despite intercoder reliability ($\text{Alpha} = 0.83$). Fourth, specific statistics may become
815 outdated quickly. Fifth, our analysis draws on published discourse rather than primary organizational data. Sixth, our analysis is conducted from a primarily Anglo-American discourse base; future cross-cultural extensions are essential to test whether the resilience paradox manifests differently under different governance regimes and labour-market structures. Seventh, the
820 rapidly evolving nature of LLM capabilities means that some specific characteristics attributed to vibe coding (e.g., the 45% vulnerability rate) may shift substantially within 12–24 months as model alignment and code-generation training mature.

6.4. Future Research Agenda

825 This study opens several avenues: (1) longitudinal case studies tracking resilience outcomes over time; (2) quantitative survey research testing our TOE model; (3) sector-specific deep dives in regulated industries; (4) environmental impact quantification of AI-assisted vs. traditional development; (5) focused investigation of the “vulnerable developer” phenomenon;
830 (6) governance framework validation through action research; (7) comparative international studies across regulatory and cultural contexts; (8) policy and regulatory analysis examining liability allocation for AI-generated code in regulated industries; and (9) business-model and macroeconomic-impact analysis estimating productivity, value-capture, and labour-market effects of
835 widespread vibe coding adoption.

As vibe coding continues its rapid diffusion, the resilience paradox identified in this study underscores the urgency: vibe coding’s extraordinary potential to accelerate digital transformation is matched by its capacity to introduce new fragilities. Organizations, policymakers, and researchers must

840 engage critically with both dimensions to ensure this paradigm shift contributes to sustainable digital resilience rather than the accumulation of hidden digital risk.

CRediT authorship contribution statement

845 **Abdulhakeem Ibrahim:** Conceptualization, Methodology, Investigation, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

Declaration of competing interest

850 The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

855 The corpus of 97 sources analysed in this study is composed of publicly available academic literature, industry reports, developer community discourse, and platform documentation. A list of included sources, the hierarchical coding framework, and AACODS quality-assessment scores are available from the corresponding author on reasonable request. No new primary data were generated.
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